

## Lesson 5-8: Determine Surface Area of Pyramids

Grade 6 Mathematics · Standard 6.GR.4

Name: \_\_\_\_\_ Date: \_\_\_\_\_

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

### Part A — Skills Check

Warm up with the lesson's core skills — one best answer each.

#### Question 1 SELECTED RESPONSE · 1 POINT

A square pyramid has a base edge of 5 in and a slant height of 7 in. What is the surface area?

- ☐ (A) 70 in<sup>2</sup>
- ☐ (B) 95 in<sup>2</sup>
- ☐ (C) 95 in<sup>3</sup>
- ☐ (D) 120 in<sup>2</sup>

#### Question 2 SELECTED RESPONSE · 1 POINT

How many lateral (triangular) faces does a square pyramid have?

- ☐ (A) 3
- ☐ (B) 4
- ☐ (C) 5
- ☐ (D) 6

#### Question 3 SELECTED RESPONSE · 1 POINT

What is the area of one triangular face with base 8 in and slant height 6 in?

- ☐ (A) 14 in<sup>2</sup>
- ☐ (B) 24 in<sup>2</sup>
- ☐ (C) 24 in<sup>3</sup>
- ☐ (D) 48 in<sup>2</sup>

**Question 4**    SELECTED RESPONSE · 1 POINT

A square pyramid has a base area of  $49 \text{ cm}^2$  and a total lateral area of  $84 \text{ cm}^2$ . What is the surface area?

- ☐ A  $49 \text{ cm}^2$
- ☐ B  $84 \text{ cm}^2$
- ☐ C  $133 \text{ cm}^2$
- ☐ D  $133 \text{ cm}^3$

**Question 5**    SELECTED RESPONSE · 1 POINT

Your first gift box is a square pyramid with a base edge of 7 in and a slant height of 9 in. How much wrapping paper (total surface area) covers it?

- ☐ A  $126 \text{ in}^2$
- ☐ B  $144 \text{ in}^2$
- ☐ C  $175 \text{ in}^2$
- ☐ D  $175 \text{ in}^3$

**Question 6**    SELECTED RESPONSE · 1 POINT

Before adding the base, you need the area of just one triangular face of a gift box with base edge 8 in and slant height 5 in. What is it?

- ☐ A  $13 \text{ in}^2$
- ☐ B  $20 \text{ in}^2$
- ☐ C  $20 \text{ in}^3$
- ☐ D  $40 \text{ in}^2$

**Part B — Test-Format Questions**

These match the state test's formats: two-part questions, select-all, and written responses.

**Question 7****TWO-PART QUESTION****Part A**

A square pyramid has a base measuring 6 m by 6 m and four triangular faces, each with a base of 6 m and a slant height of 5 m. What is its surface area?

*Fill in the bubble next to the one best answer.*

- ☐ (A) 36 square meters
- ☐ (B) 60 square meters
- ☐ (C) 96 square meters
- ☐ (D) 156 square meters

**Part B**

*Answer Part A before Part B — Part B asks about your reasoning.*

Why does a pyramid's surface area use the slant height rather than the vertical height?

- ☐ (A) The slant height is the height of each triangular face, measured along the face itself.
- ☐ (B) The vertical height is always longer than the slant height.
- ☐ (C) The slant height is easier to measure.
- ☐ (D) Pyramids do not have a vertical height.

**Question 8****SELECT ALL THAT APPLY**

Select ALL statements that are true about the surface area of a pyramid:

*Mark the box next to EVERY answer that is true. More than one is correct.*

- ☐ (A) Surface area = base area + the areas of all lateral faces.
- ☐ (B) A square pyramid has 4 triangular lateral faces.
- ☐ (C) Each lateral face area is  $\frac{1}{2} \times \text{base of the triangle} \times \text{slant height}$ .
- ☐ (D) The vertical height is used to find each triangular face area.
- ☐ (E) A pyramid's net shows the base and every lateral face laid flat.

**Part C — Show Your Thinking**

Answer in writing, the way the state test's constructed responses work.

**Question 9****WRITTEN RESPONSE · 2 POINTS**

Two pyramids have the same base area ( $64 \text{ in}^2$ ) but different slant heights. Pyramid A has slant height 6 in and Pyramid B has slant height 10 in. How much more surface area does Pyramid B have? Why does slant height affect surface area but NOT base area?

*Write your answer in complete sentences. Show or explain your mathematical thinking.*

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When you finish, check your reasoning with your teacher or family. Your teacher has the scoring guide for the written responses.